File Copy Implementation

1. Team Members and Responsibility

As this is an individual assignment, all tasks were performed by a single member:

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2. Project Overview

This project is a low-level file utility implemented in **C** using **POSIX system calls**. It demonstrates the direct interaction between user-level applications and the operating system kernel for file I/O operations.

File Copy Logic

The program operates through the following steps:

- **User Prompting**: Requests the source and destination filenames from the user.
- **File Descriptors**: Opens the source file in read-only mode and the destination file with creation/truncation flags.
- **Buffered Transfer**: Performs a byte-by-byte copy using a 1024-byte buffer to optimize read/write performance.

3. Features

- **Direct System Calls**: Utilizes `open()`, `read()`, `write()`, and `close()` for low-level I/O.
- **Robust Error Handling**: Checks for file existence, directory permissions, and potential write errors.
- **POSIX Compliant**: Developed and tested on macOS (Darwin), ensuring compatibility with most Unix-like systems.

4. Compilation and Configuration Instructions

Environment

- **OS**: Developed on **macOS (Apple Silicon)**; compatible with Unix-like systems (Linux/BSD).
- **Requirements**: `gcc` or `clang` compiler.

Build the Project

Enter the project folder and use one of the following command to compile:

Method A. Using Makefile(Recommended):

```
make
```

Method B. Direclty command to complie

```
gcc FileCopy.c -o FileCopy
```

Run the Program

Execute the binary and follow the on-screen prompts:

```
./FileCopy
# Enter source file name: source.txt
# Enter destination file name: dest.txt
```

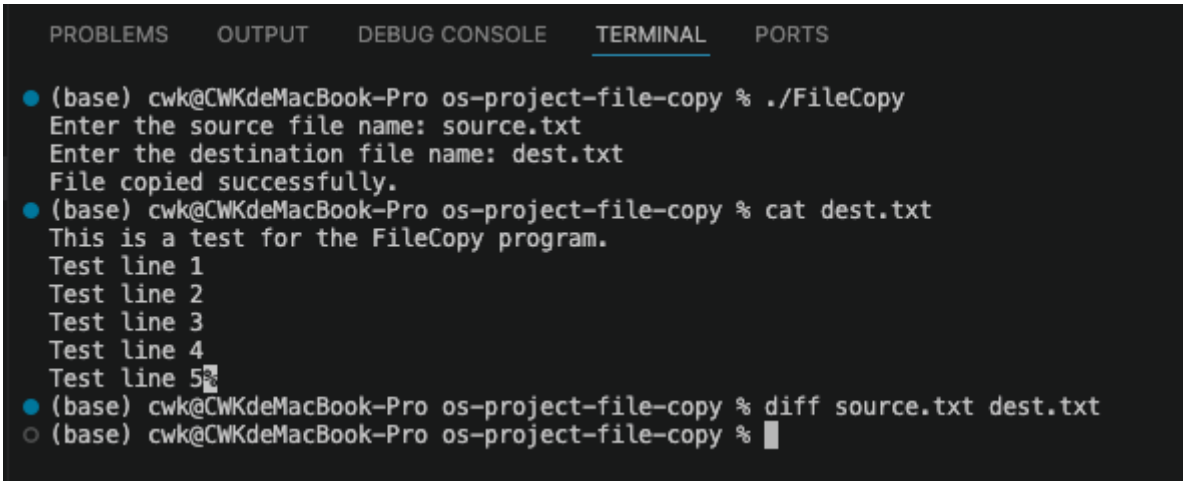
Verification

To ensure the copy is identical to the source, use the diff command:

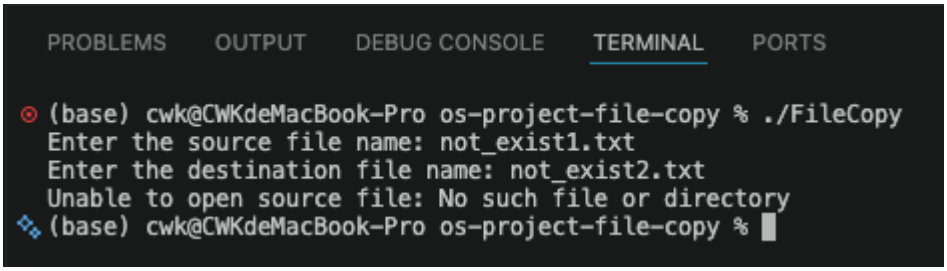
```
diff source.txt dest.txt
# No output means the files are identical.
```

5. Test Results (System Trace & Error Handling)

a. Basic Feature: Successful Copy



b. Error Handling: Source File Not Found



c. Error Handling: Permission Denied

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

(base) cwk@CWKdeMacBook-Pro os-project-file-copy % ./FileCopy
Enter the source file name: secret.txt
Enter the destination file name: dest.txt
Unable to open source file: Permission denied
(base) cwk@CWKdeMacBook-Pro os-project-file-copy %
```

d. System Trace: dtruss via SIP

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

(base) cwk@CWKdeMacBook-Pro os-project-file-copy % sudo dtruss ./FileCopy
dtrace: system integrity protection is on, some features will not be available

SYSCALL(args)             = return
Enter the source file name: source.txt
Enter the destination file name: dest.txt
File copied successfully.
```

Observation of kernel-level system calls. The `dtruss` output is restricted by macOS **System Integrity Protection (SIP)** on modern macOS for security reasons.

However, the program's logic is verified via `diff` and binary comparison. To view traces, ensure the binary is not restricted by SIP or run in a sandbox-relaxed environment.

6. File Structure

- `FileCopy.c`: Core source code (POSIX I/O logic).
- `Makefile`: Automation script for compilation and cleanup.
- `README.md`: Project documentation.
- `snapshot`: Directory containing execution screenshots and test results.

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